Pranshu Tripathi

Software Engineer

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SUMMARY

Software Development Engineer with 2+ years of experience across full-stack development, system design, and AI integrations. Proven expertise in C++, Python, FastAPI, and Kubernetes, with deep knowledge of design patterns, CI/CD, and microservices. Recognized by executive leadership for architecting the LLM-based Maestro Assistant and leading the development team to drive the project from prototype to global launch. Delivered a 300-400x performance boost in molecular calculations using a custom C++ library. Actively seeking challenging SDE roles to build scalable, high-performance systems in cutting-edge tech environments.

EXPERIENCE

Schrödinger

Senior Developer I *January 2026 - Present*

- Project lead of Maestro Assistant.
- Developed a Semantic Panel Search prototype using vector embeddings to replace legacy algorithmic search, successfully validating the architecture for production deployment
- Collaborating on architecture design of Notification and Authentication services for Maestro.

Developer II *July 2024 - December 2025*

- Engineered the "Maestro Assistant" service into a single proxy, implementing CI/CD pipelines and telemetry to support 100+ daily production requests. Received recognition from company leadership and customers like BMS, GSK.
- Implemented non-bonded interaction calculation library; a modular C++ library with caching. It reduced the calculation time by 300-400x; cut 10-min tasks to 30s.
- Mentored an engineering intern on the Project Table Calculator, implementing daily code reviews and enforcing scalable design patterns to ensure the successful delivery of major project components.

Developer I *June 2023 - June 2024*

- Refactored 2000+ lines of legacy code using Factory and Strategy patterns, improving extensibility for handling diverse Job Server outputs.
- Enhanced export functionality for new movie presets (e.g., View Ligand Docking) from Maestro to PyMOL, decreasing manual visualization setup time by 60% and enhancing automated workflow efficiency.

Intern - Software Development *January 2023 - May 2023*

- Created an adapter for Schrödinger's structure class that removed the m2io dependency and enabled a property-writing mechanism with 3x greater flexibility, gaining recognition from Scientific Infrastructure team.
- Implemented a robust Find Toolbar widget following MVC pattern and command pattern with python and C++ inter communication.

Intern - Software Development *May 2022 - July 2022*

- Delivered a robust preset management feature in 2 months—30% faster thanaverage—using Factory and Adapter design patterns.
- Automated vertex AI steps for Maestro Assistant using Kubernetes and GitHub, enhancing the efficiency of the resume update process.
- Validated Phase 6 by generating a reviewed GitHub PR from a Kubernetes worker after compiling the Resume repository with latexmk, ensuring the accuracy of resume updates.
- Implemented a repair flow for resume updates by allowing the Kubernetes worker to request corrected resume update plans based on LaTeX safety feedback, improving the reliability of the resume generation process.
- Automated vertex AI steps for Maestro Assistant using Kubernetes and GitHub, enhancing the efficiency of the resume update process.
- Validated Phase 6 by generating a reviewed GitHub PR from a Kubernetes worker after compiling the Resume repository with latexmk, ensuring the accuracy of resume updates.
- Implemented a repair flow for resume updates by allowing the Kubernetes worker to request corrected resume update plans based on LaTeX safety feedback, improving the reliability of the resume generation process.

PROJECTS

EventHive

Deployed a ticketing platform on GKE handling 3,000+ RPS, publishing NPM packages to streamline microservice integration across 5 services

PlotCortex

A place to semantically search movies and TV shows in IMDB movies dataset

CarbonHero

Built a leaderboard-based Flutter app for GDSC Solution Challenge 2021 to track user carbon footprint, integrating Firebase for real-time updates and auth

EDUCATION

IIIT Allahabad (B.Tech)

Electronics and Communication

Aug 2019 - May 2023

- **GPA: 9.16/10**
- **Batch Rank: 4**
- **Award:** Merit-cum-Means Scholarship for Academic Excellence 2020, 2021, 2022

AWARDS

- Runner-up in Schrödinger Hackathon 2024
- Appreciation from CEO, CTO for Maestro Assistant Project
- Silver Medal - Football, Badminton & Table Tennis at Schrödinger Sports Meet

SKILLS

Programming Languages

- C++, Python, Java, JavaScript, TypeScript, SQL, Shell Scripting, Terraform

Framework & Libraries

- Spring, Express.js, React, Next.js, Qt, Hibernate, PostgreSQL, SIP (Python-C++), PyQt6, BULLJS

Cloud & DevOps

- Docker, Kubernetes (K8s), GitHub Workflows, Travis CI, AWS (Elastic Beanstalk, RDS, ElastiCache), GCP, GKE, G Cloud SQL, NATs Streaming

Generative AI Technologies

- LiteLLM, Vertex AI, Hugging Face, Qdrant Vector DB, LangChain, DSPY, MCP, RAG

Additional Interests

- CNNs, LSTMs, Transformers, Graph DB, IoT Systems, Classification Models, Speech to Text recognition

VOLUNTEERING

- Coordinator - IoT Wing, IIIT Allahabad
- Senior Member - Literature Society, IIIT Allahabad

LANGUAGES

English, Hindi, Marathi, German